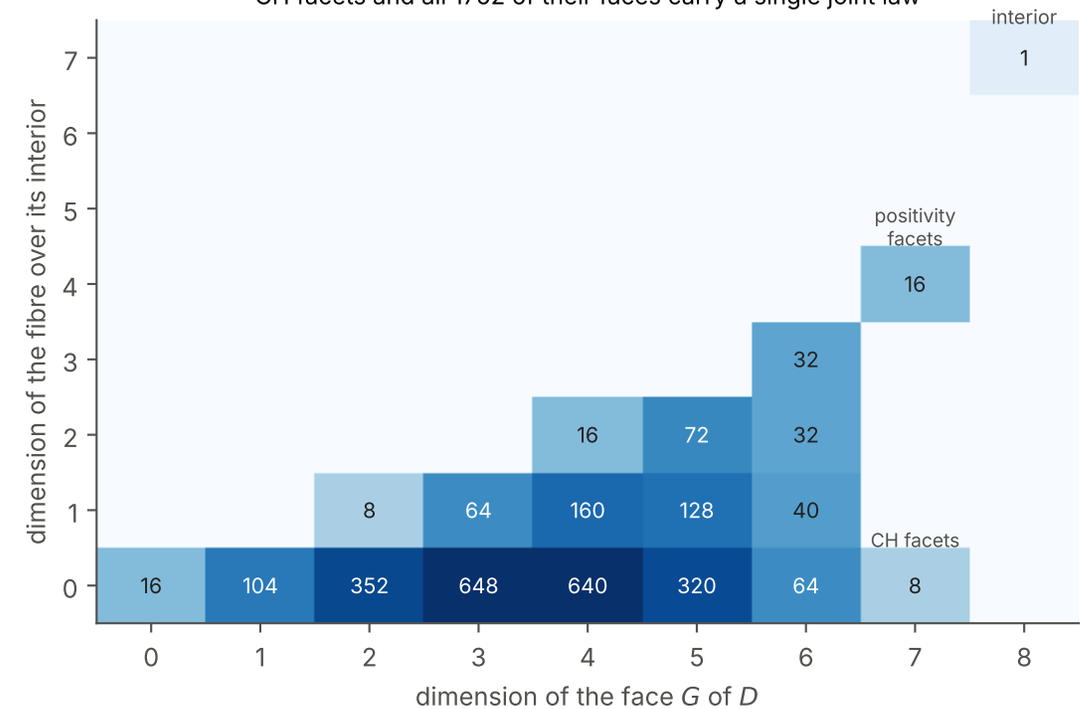
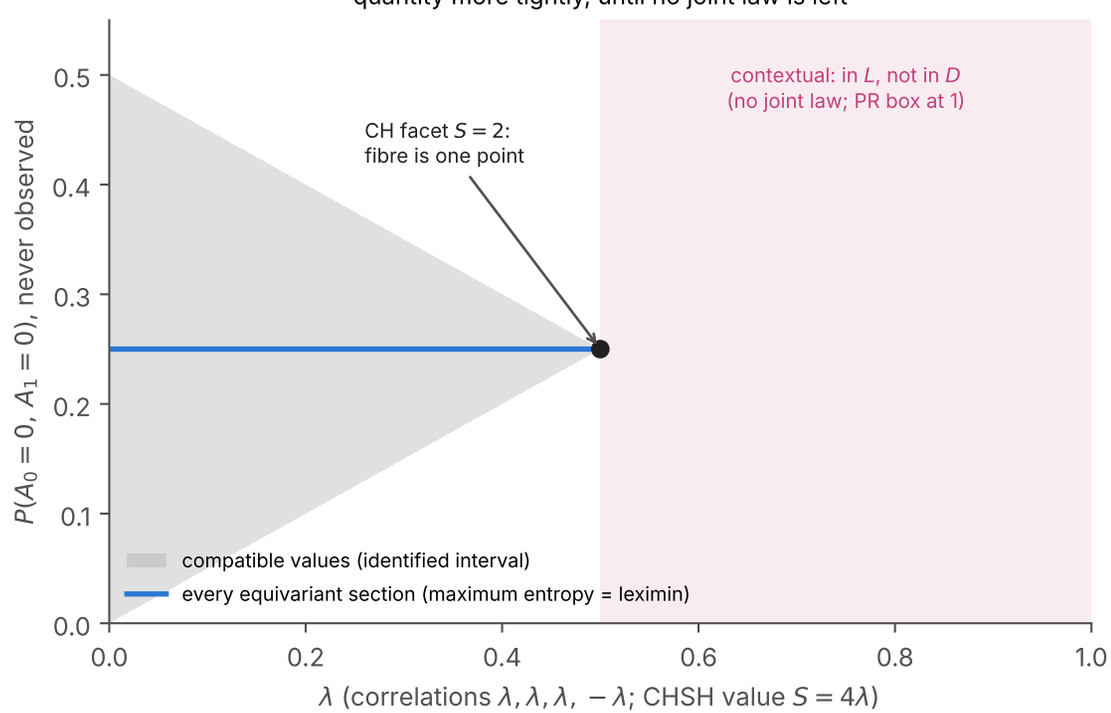
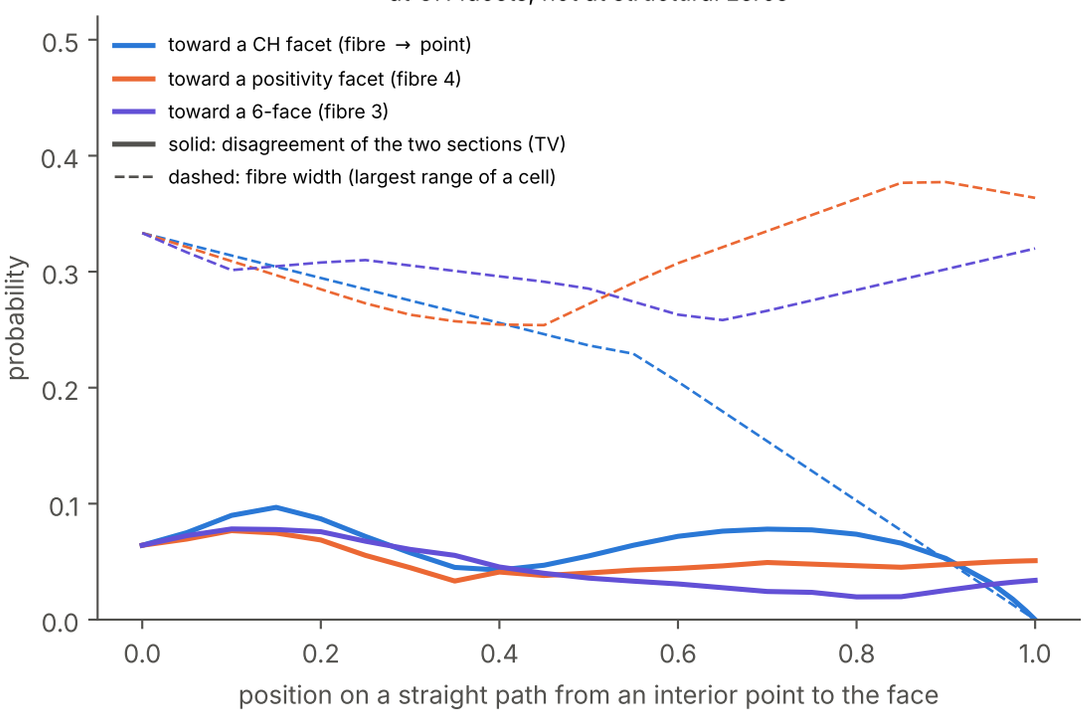
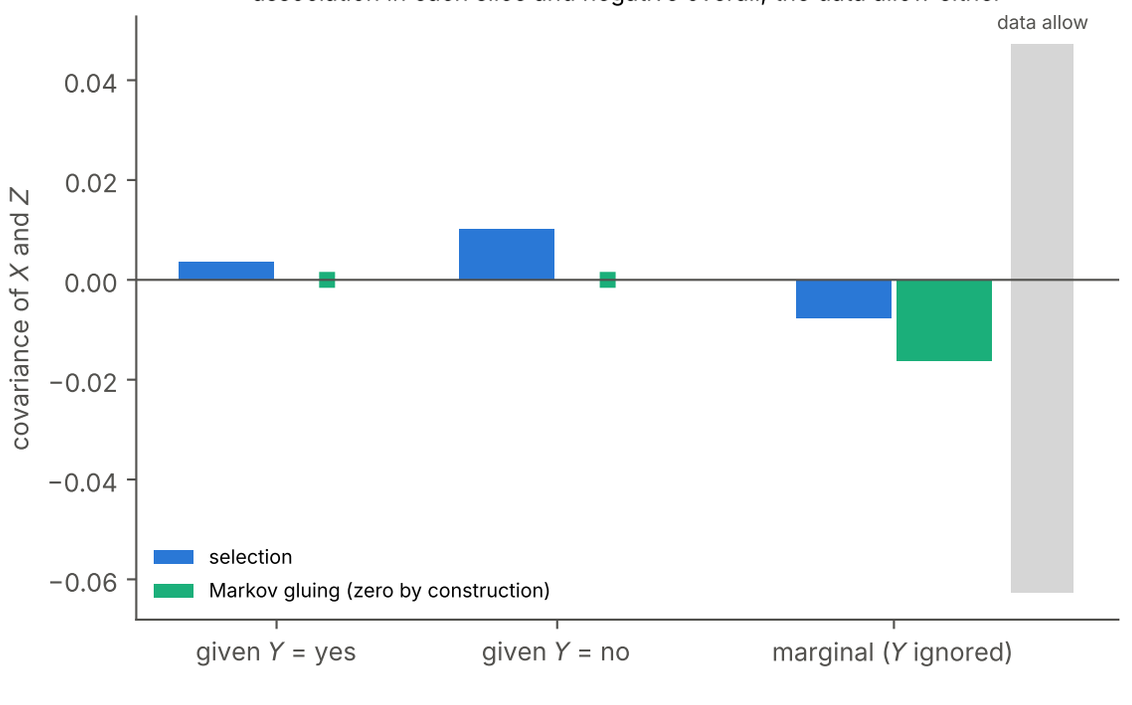

(a) The 4-cycle: fibre dimension = |V(G)| − 1 − dim G over each face.
CH facets and all 1752 of their faces carry a single joint law
interior
1
positivity facets
16
32
16
72
32
8
64
160
128
40
CH facets
16
104
352
648
640
320
64
8
dimension of the fibre over its interior
dimension of the face G of D
0
1
2
3
4
5
6
7
8


(b) The symmetric CHSH path: the data pin the unobserved quantity more tightly, until no joint law is left
contextual: in L, not in D
(no joint law; PR box at 1)
CH facet S = 2:
fibre is one point
compatible values (identified interval)
every equivariant section (maximum entropy = leximin)
P(A0 = 0, A1 = 0), never observed
λ (correlations λ, λ, λ, −λ; CHSH value S = 4λ)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.8
1.0


(c) Rules are forced to agree only where the fibre collapses:
at CH facets, not at structural zeros
toward a CH facet (fibre → point)
toward a positivity facet (fibre 4)
toward a 6-face (fibre 3)
solid: disagreement of the two sections (TV)
dashed: fibre width (largest range of a cell)
probability
position on a straight path from an interior point to the face
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.8
1.0


(d) Simpson's reversal made by a rule: selection gives positive
association in each slice and negative overall; the data allow either
data allow
covariance of X and Z
selection
Markov gluing (zero by construction)
given Y = yes
given Y = no
marginal (Y ignored)
0.04
0.02
0.00
−0.02
−0.04
−0.06